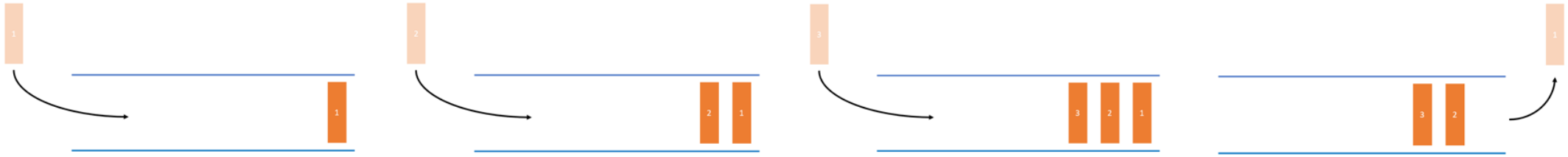
Lecture 15 - Binary Heaps and Priority Queues

Foundations of Computer Science (4CC505)

Dr Sam O'Neill

Queue



enqueue()

Item 1

enqueue()

Item 2

enqueue()

Item 3

dequeue()

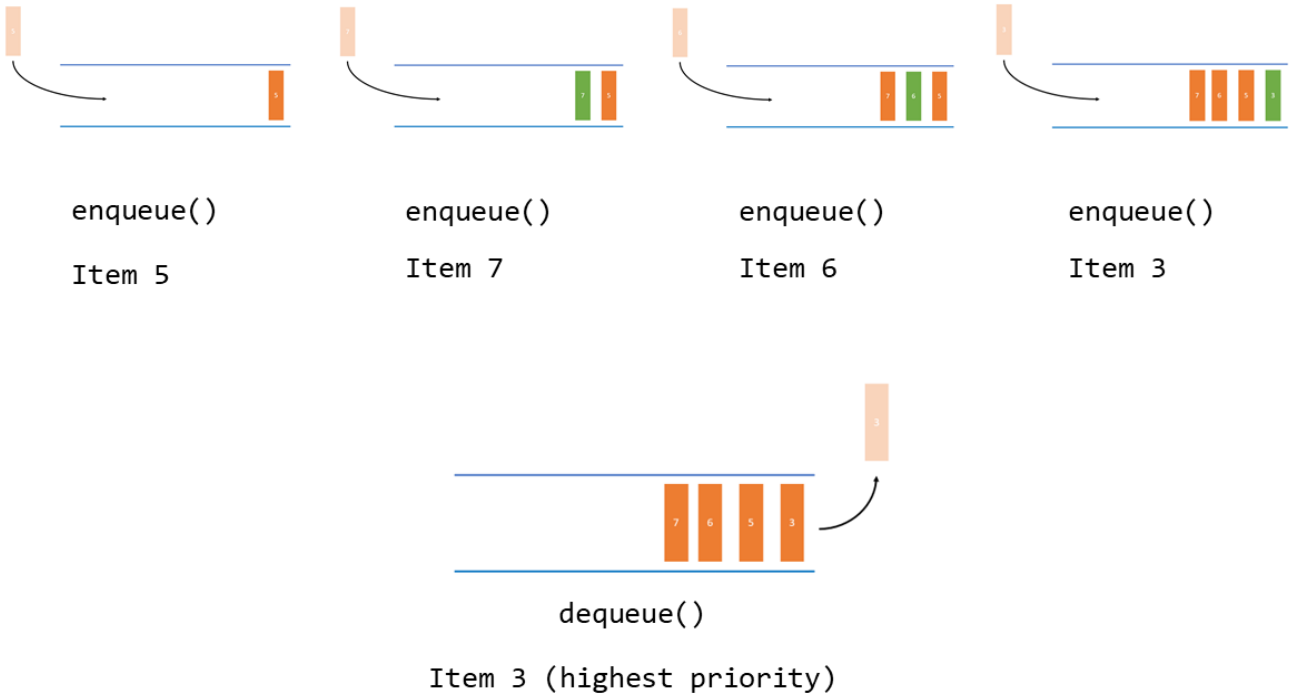
Item 1

Queue (ADT)

Operation	Description
`enqueue(x)`	Add item x to the end of the queue
`dequeue()`	Remove the item at the front of the queue and return it
`is_empty()`	Check if the queue is empty
`is_full()`	Check if the queue is full
`peek()`	View the item at the front of the queue without removal

Priority Queue

Queue in which items are ordered by priority.



Air Traffic Control Example

- You have scheduled aircraft **X/Y** to land in the next $3/6$ minutes, respectively.
- Both have enough fuel for at least the next 15 minutes and both are just 2 minutes away from your airport.
- You observe that your airport runway is clear at the moment.

```
🛩️(Y)
          🛩️(X)
                    (T)
                  <---->
                   |||
                   |||
_________________|||_________________
```

Visualgo.net

Air Traffic Control Example

- Suddenly, you receive an urgent SOS message that another aircraft **Z** is running out of fuel and request to land soon.
- The pilot of aircraft **Z** estimates that he only have 3 minutes of flying time and also approximately 3 minutes away from your airport...

```
✈️(Z)
                ✈️(Y)
                              ✈️(X)
                                          (T)
                                        <--->
                                         |||
                                         |||
                                 ________|||_______________
```

Air Traffic Control Example

Do you?

- Let aircraft **Z** land first and hold **X** and **Y**
- Stick with the original plan

Priority Queue (ADT)

We will look at the max priority queue, but we can do a min priority queue.

Now each item `x` in the queue has an associated `key` (commonly the value of `x`).

Operation	Description
`insert(x)`	Insert item `x` based on it's priority (`key`) (like `enqueue(x)`)
`extract_max()`	Return the item with the highest priority (`key`) (like `dequeue()`)
`is_empty()`	Check if the queue is empty
`is_full()`	Check if the queue is full
`peek()`	View the item at the front of the queue without removal

Using Python's `queue` Module

An example using the `PriorityQueue` class in Python.

NOTE - This returns items based on the minimum!

```
# https://builtin.com/data-science/priority-queues-in-python
from queue import PriorityQueue
customers = PriorityQueue() #we initialise the PQ class instead of using a function to operate upon a list.
customers.put((2, "Harry", "123456")) # insert()
customers.put((3, "Charles")) # insert()
customers.put((1, "Riya")) # insert()
customers.put((2, "Stacy")) # insert()
customers.put((2, "Abid")) # insert()
while not customers.empty(): # corrected error on the webpage above
     print(customers.get()) # extract_min()

# prints in the following order - (1, 'Riya'), (2, 'Abid'), (2, 'Harry', '123456'), (2, 'Stacy'),  (3, 'Charles')
```

So, `(2, 'Abid') < (2, 'Harry', '123456') < (2, 'Stacy')` because Python first sorts on the `2`s and then the names `'Harry'`, `'Stacy'` and `'Abid'`.

My Implementation

NOTE - This returns items based on the maximum!

- This uses the classic version from ***CLRS - Introduction to algorithms*** (see references).

```
from priority_queue import PriorityQueueADT

customers = PriorityQueueADT()

customers.insert((2, "Harry", "123456")) # insert()
customers.insert((3, "Charles")) # insert()
customers.insert((1, "Riya")) # insert()
customers.insert((2, "Stacy")) # insert()
customers.insert((2, "Abid")) # insert()

while not customers.is_empty(): # corrected error on the webpage above
  item = customers.extract_max() # extract_max()
  print(f"Key: {item.key}, Value: {item.value}")

# prints in the following order - (3, 'Charles'), (2, 'Stacy'), (2, 'Harry', '123456'), (2, 'Abid'), (1, 'Riya')
```

Using a Binary Heap to Implement a Priority Queue

- We are going to use a **binary heap** data structure to implement a **priority queue**.
- Spoiler - the time complexities are as follows.

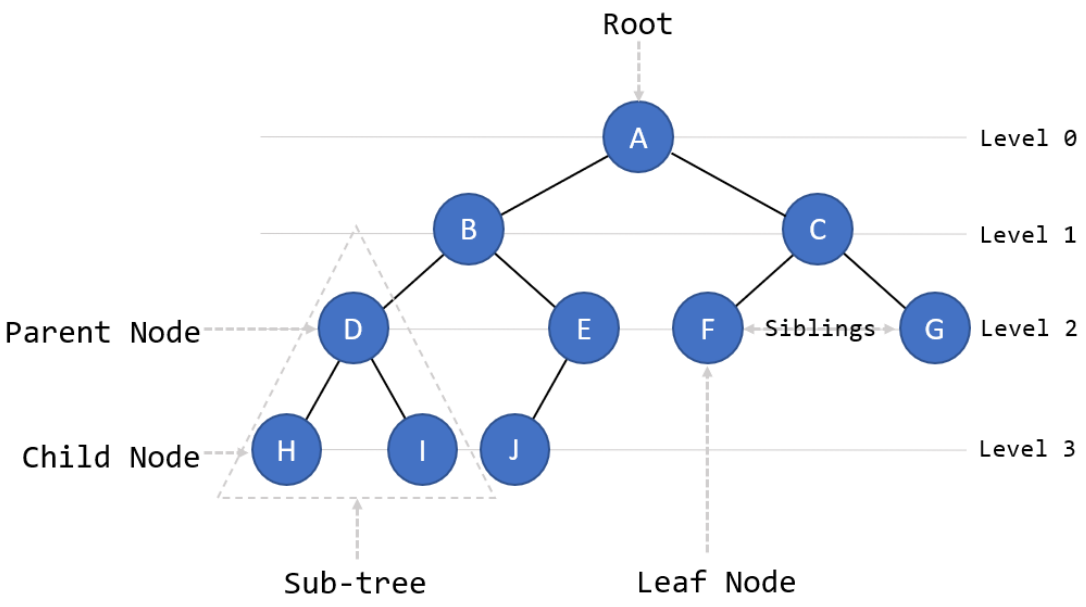
Operation	Average-case Cost	Worst-case Cost
`insert(x)`	$O(1)$	$O(\log(n))$
`extract_max()`	$O(\log(n))$	$O(\log(n))$
`peek()`	$O(1)$	$O(1)$

Discussion on time complexity

Binary Tree

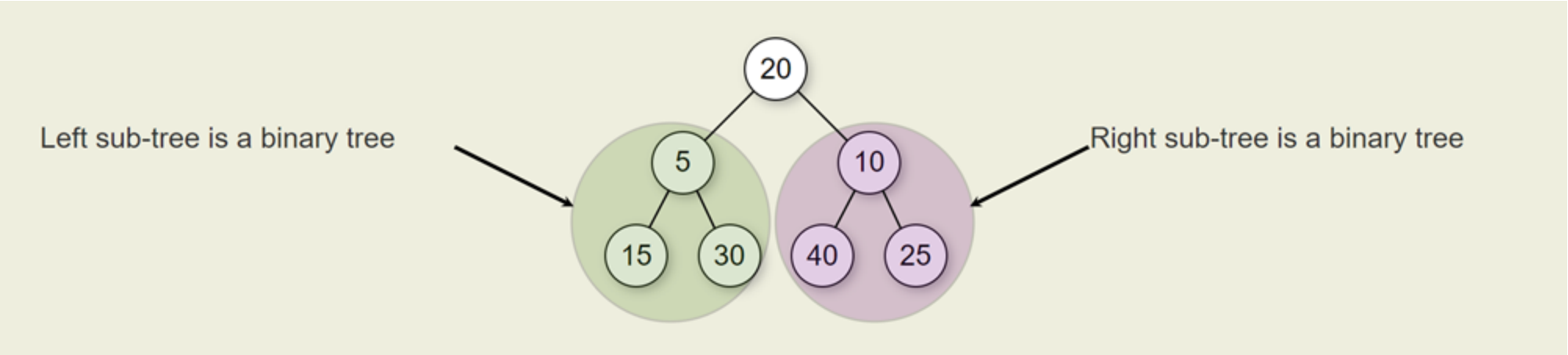
A binary tree has:

- a root node
- each node has at most two children
  - a left child
  - a right child



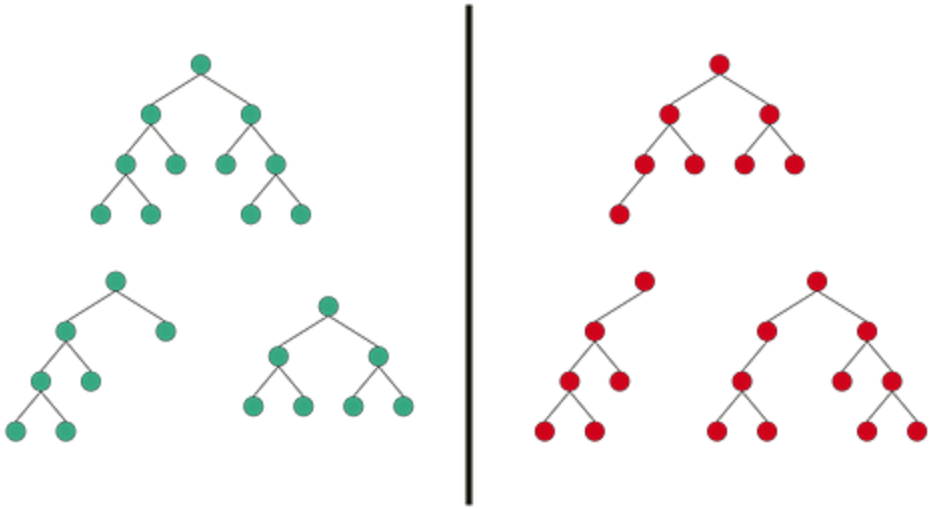
Sub-trees

- The root pointer points to the topmost node in the tree.
- The left and right pointers recursively point to smaller subtrees on either side.
- A **NULL** pointer represents a subtree with no children.



Full Binary Tree

- **Full** - Every node has 0 or 2 children

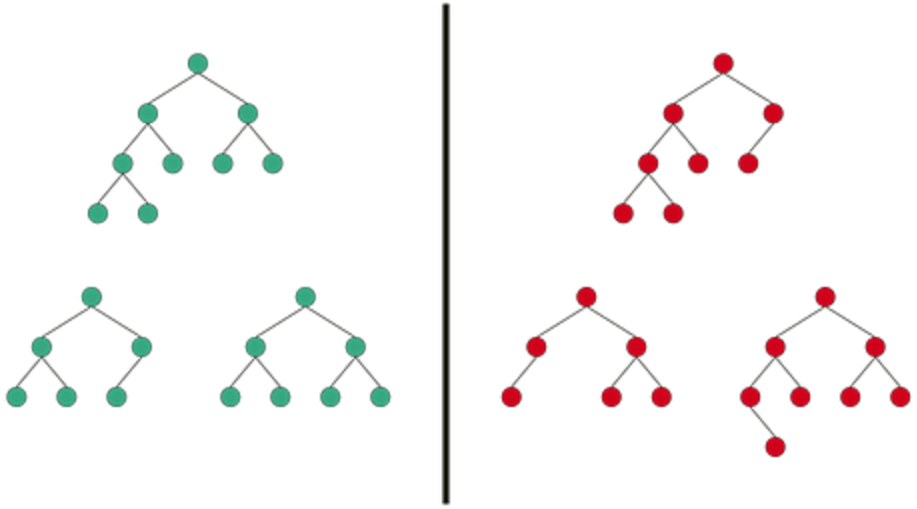


https://towardsdatascience.com/5-types-of-binary-tree-with-cool-illustrations-9b335c430254

Green examples are **full** binary trees.

Complete Binary Tree

- **Complete** - Every level, except possibly the last, is completely filled, and all nodes in the last level are as far left as possible.
- Height is $\lfloor \log(n) \rfloor$ (n is number of nodes).



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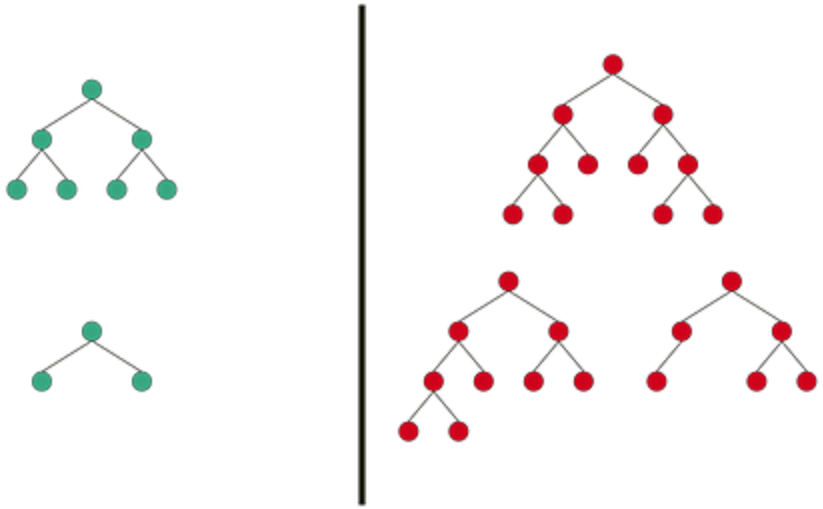
Green examples are **complete** binary trees.

Perfect Binary Tree

- **Perfect** - All interior nodes have two children and all leaves have the same depth or same level.
- Height is $\lfloor \log(n) \rfloor$ (n is number of nodes).



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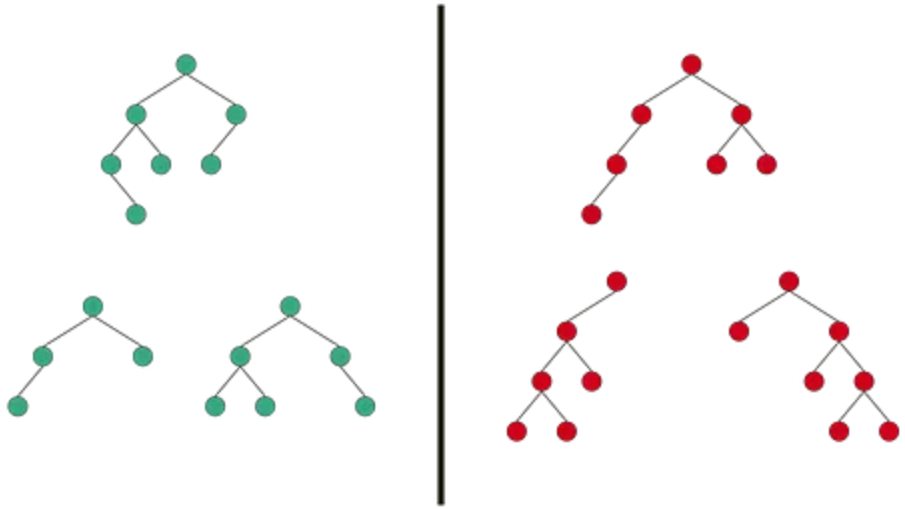
Green examples are **perfect** binary trees.

Balanced Binary Tree

- **Balanced** - Left and right subtrees of every node differ in height by no more than 1



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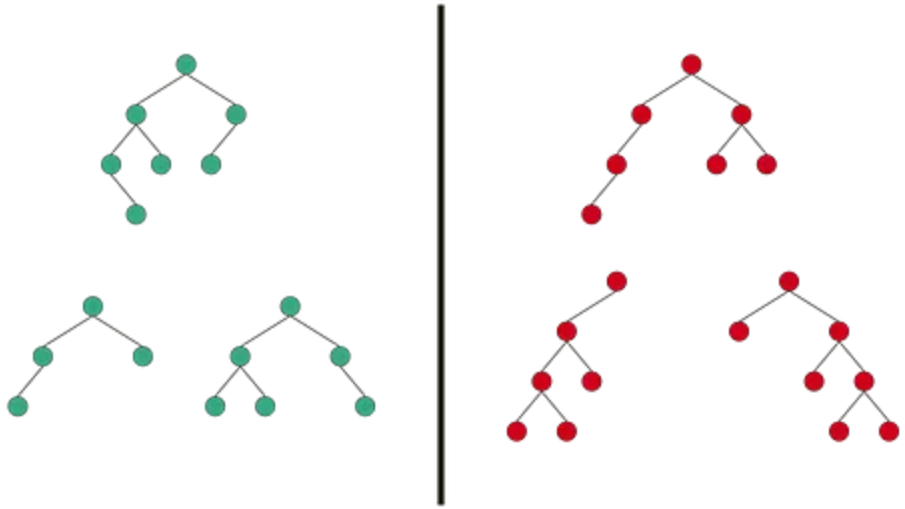
Green examples are **balanced** binary trees.

Degenerate (Pathological) Binary Tree

- **Degenerate** - Each parent node has only one associated child node



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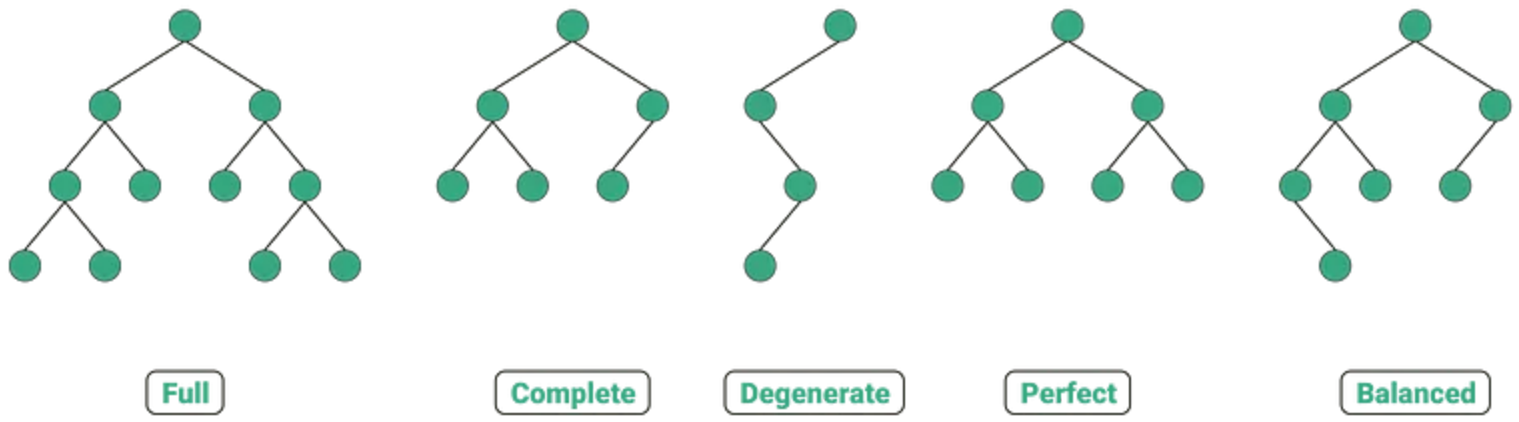
Green examples are **degenerate** binary trees.

Binary Tree Types



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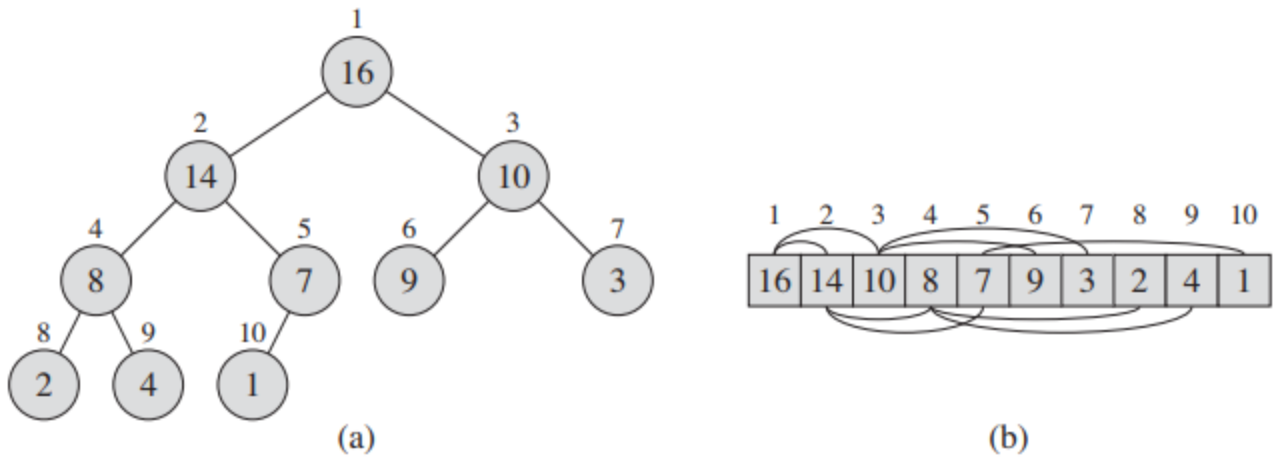
Binary Heap

- **Data Structure** we can use to efficiently **implement a priority queue**.
- Either:
  - **Max-heap**
  - **Min-heap**
- A **complete** binary tree.
- Every node satisfies the **heap property**.

Max-heap Example



A max-heap viewed as **(a)** a binary tree and **(b)** an array.

- The root node has the biggest value.
- Every node satisfies the **max-heap property**.
  - the value of each node is less than or equal than it's parents.

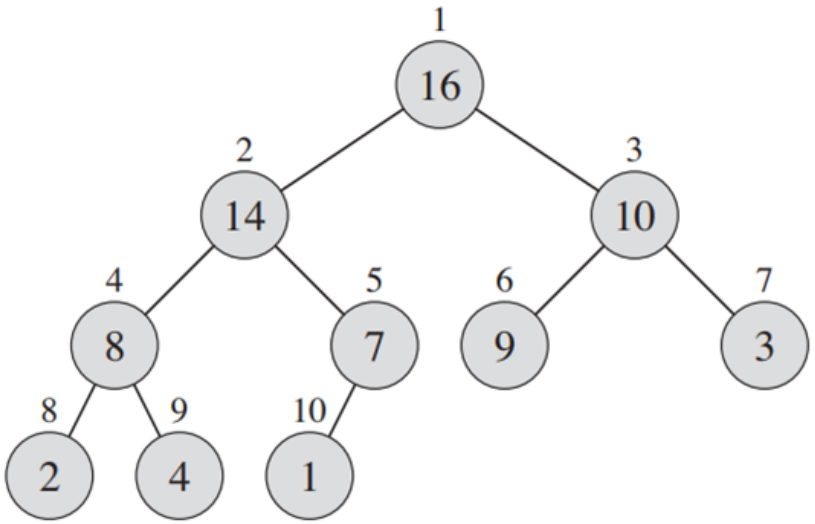
Does the binary tree in (a) satisfy the **max-heap property**?

Parent, Left-child and Right-child

```
PARENT(i)
    return ⌊i/2⌋

LEFT(i)
    return 2i

RIGHT(i)
    return 2i + 1
```



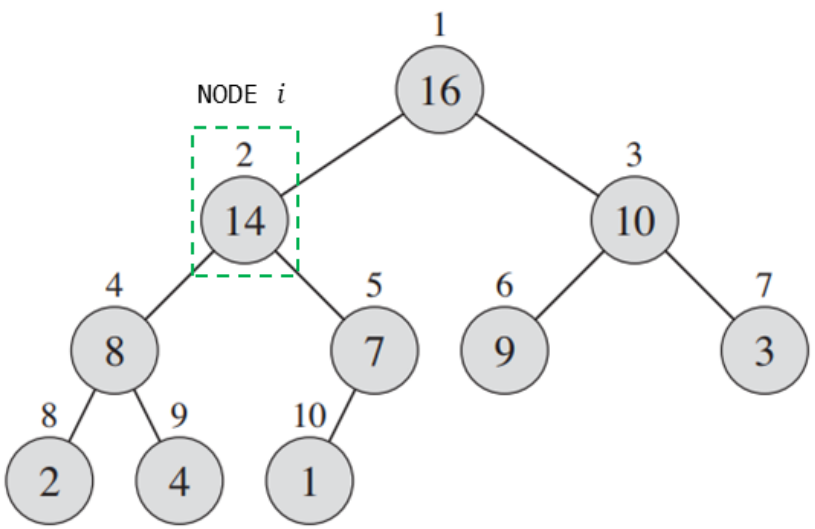
These are highlighted on the next 3 slides…

Parent, Left-child and Right-child

```
PARENT(i)
    return ⌊i/2⌋

LEFT(i)
    return 2i

RIGHT(i)
    return 2i + 1
```

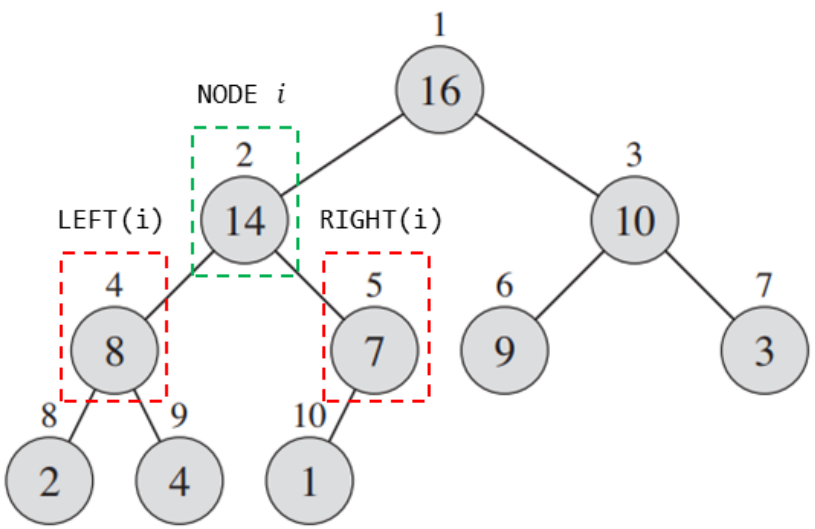


Parent, Left-child and Right-child

PARENT(i)
    return $\lfloor i/2 \rfloor$

LEFT(i)
    return $2i$

RIGHT(i)
    return $2i + 1$



Parent, Left-child and Right-child

PARENT(i)
    return $\lfloor i/2 \rfloor$

LEFT(i)
    return $2i$

RIGHT(i)
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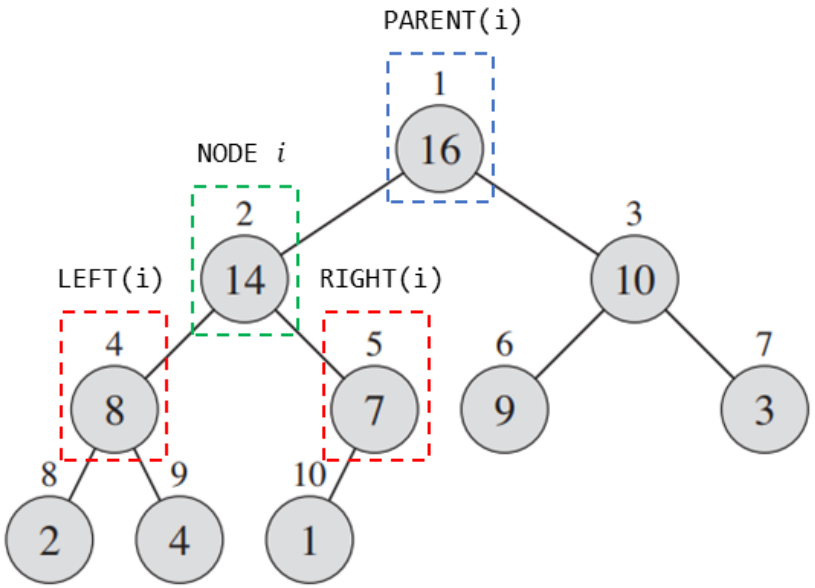
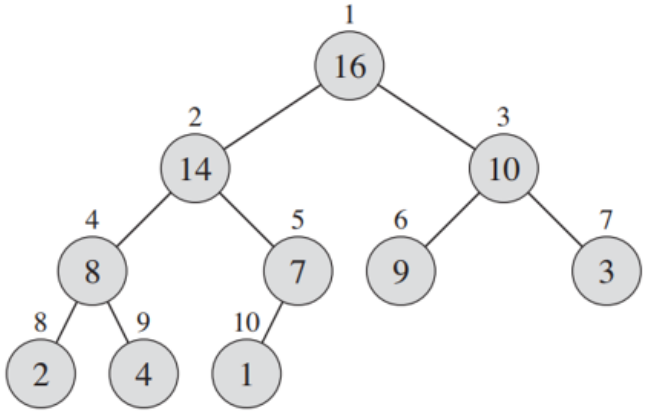


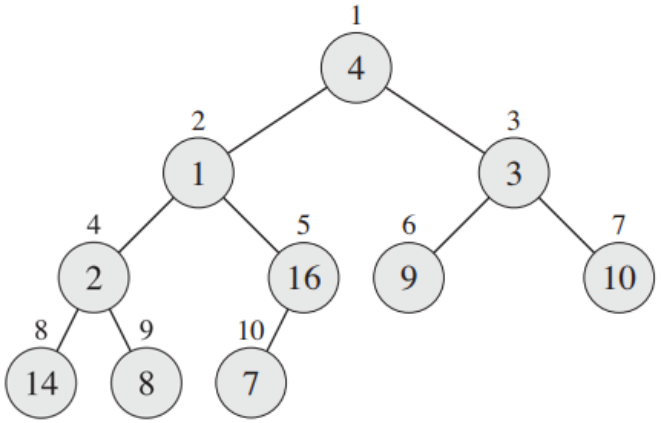
Heap Property

- **Heap Property** (either Max-Heap Property or Min-Heap Property).
- e.g. **Max-heap Property**.
  - For every node with index i.
    - `A[parent(i)] >= A[i]`

      That is, the value of each node is less than or equal than it's parents.



Satisfies max-heap property



Does not satisfy max-heap property

TASK

Visit visualgo.net (Binary Heap)

- Create a binary heap
- Insert an item (Keep doing this until you understand what is happening)
- Extract Max (Keep doing this until you understand what is happening)

Keep an eye on the **max-heap property** for all nodes in the heap.

Maintaining the Max Heap Property

From now on we will talk about max-heaps.

When we either:

- extract the max item (root node)
- insert a new item

We must reorder the nodes so that the **max-heap property** is maintained.

Sift Down

Sifting down on a subtree rooted at index i.

Determine the largest of the nodes:

- `A[i]` (parent)
- `A[left(i)]` (left-child)
- `A[right(i)]` (right-child).

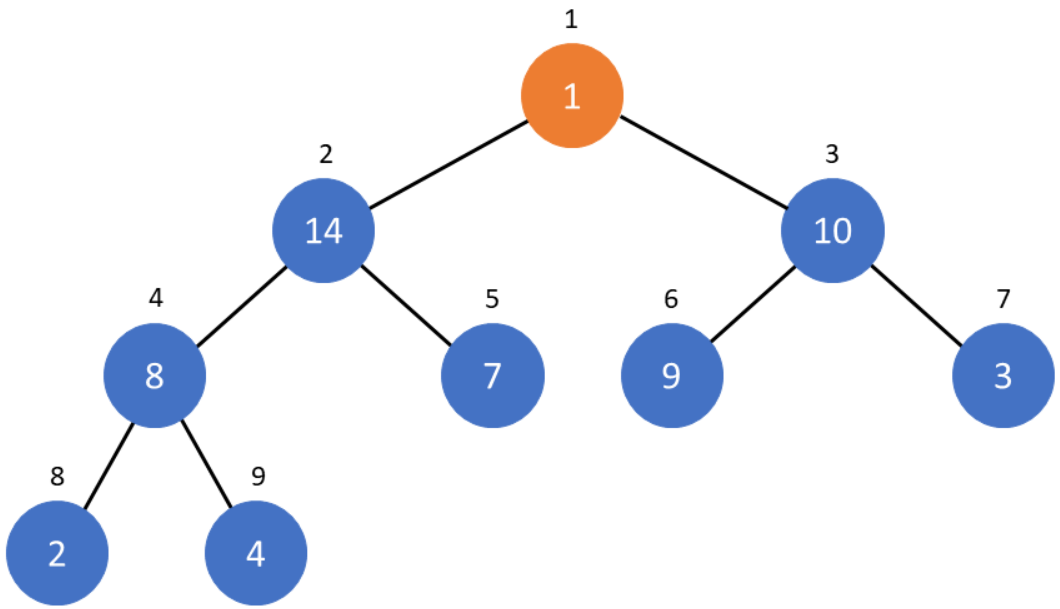
and store the index as `largest`.

1. If `A[i]` is `largest` then subtree satisfies the **max-heap property**.
2. Otherwise, exchange the `A[i]` and `A[largest]`.

Example of Sift Down

Animation, please move to the next slide...

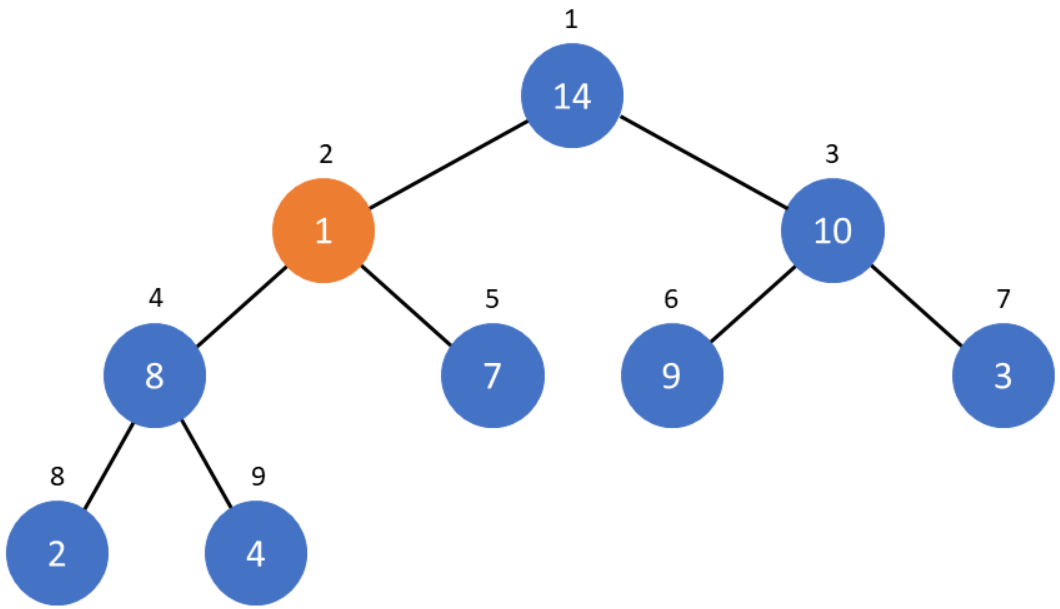
- Largest of the nodes indexed at 1, 2 and 3 is the node at index 2 with value 14.
- Therefore sift down - Swap nodes indexed at 1 (parent) and 2 (left-child).



Example of Sift Down

Animation, please move to the next slide...

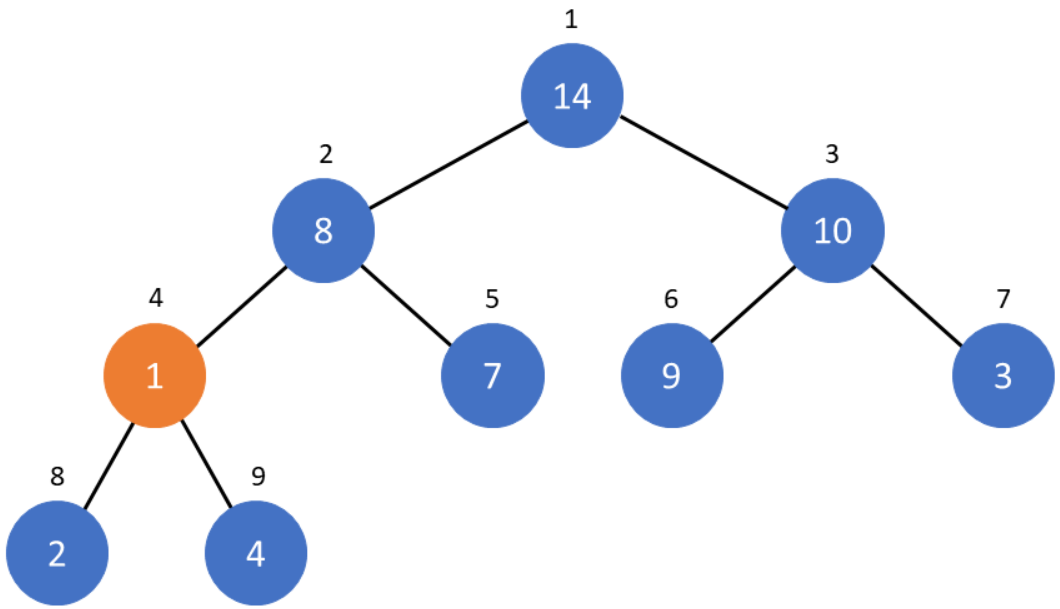
- Largest of the nodes indexed at 2, 4 and 5 is the node at index 4 with value 8.
- Therefore sift down - Swap nodes indexed at 2 (parent) and 4 (left-child).



Example of Sift Down

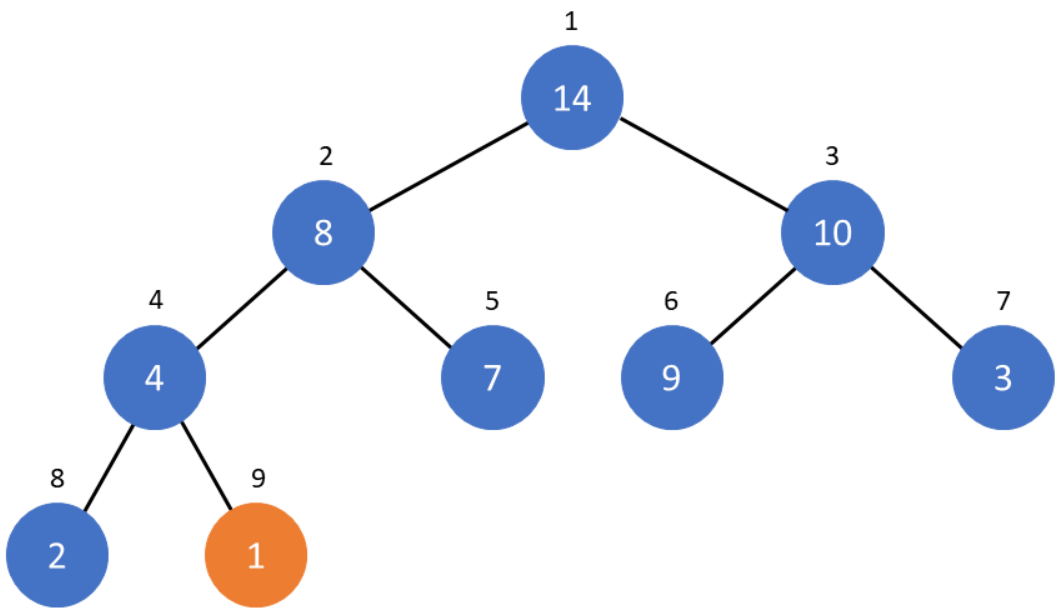
Animation, please move to the next slide...

- Largest of the nodes indexed at 4, 8 and 9 is the node at index 9 with value 4.
- Therefore sift down - Swap nodes indexed at 4 (parent) and 9 (right-child).



Example of Sift Down

- Node with value 1 is now a leaf node.



Extracting Max (`extract_max()`)

1. Extract the root node.
2. Make the last node in the tree the root node
3. Sift down the root node until at correct position

Please try extracting max on Visualgo.net

This operation takes $O(log(n))$.

Why? In the worst-case we have to sift the new root node all the way down to the leaf nodes. Height of complete tree is $\lfloor \log(n) \rfloor$.

Sift Up

Sifting up a node at index i.

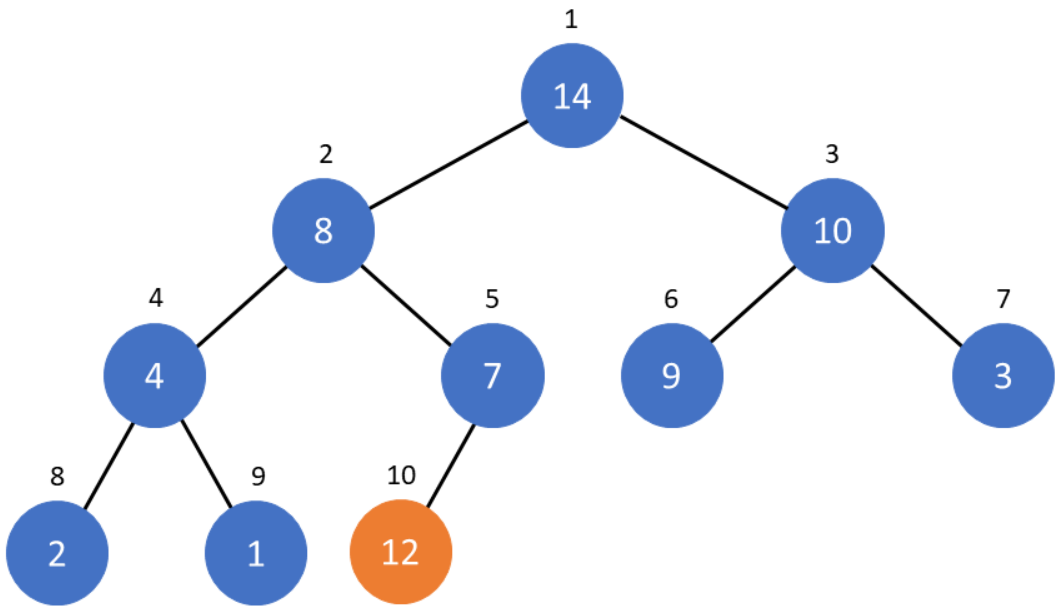
1. If node at index i is larger than its' parent, i.e. `A[i] > A[parent[i]]`, then exchange.

This is normally repeated until the above is not true.

Example of Sift Up

Animation, please move to the next slide...

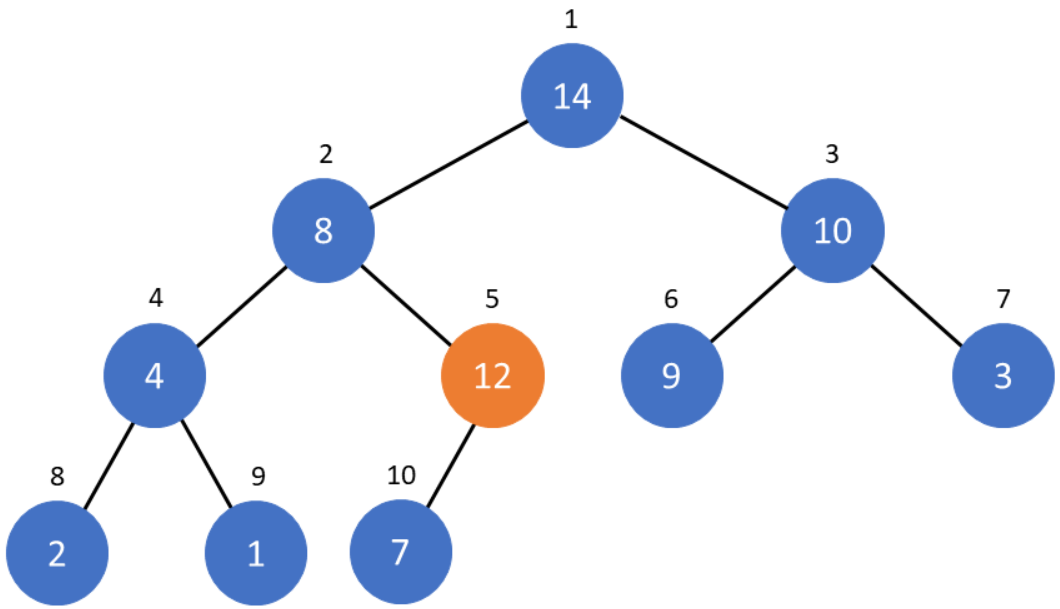
- Node indexed at 10 has a value higher than it's parent (node at index 5).
- Swap index 10 and index 5.



Example of Sift Up

Animation, please move to the next slide...

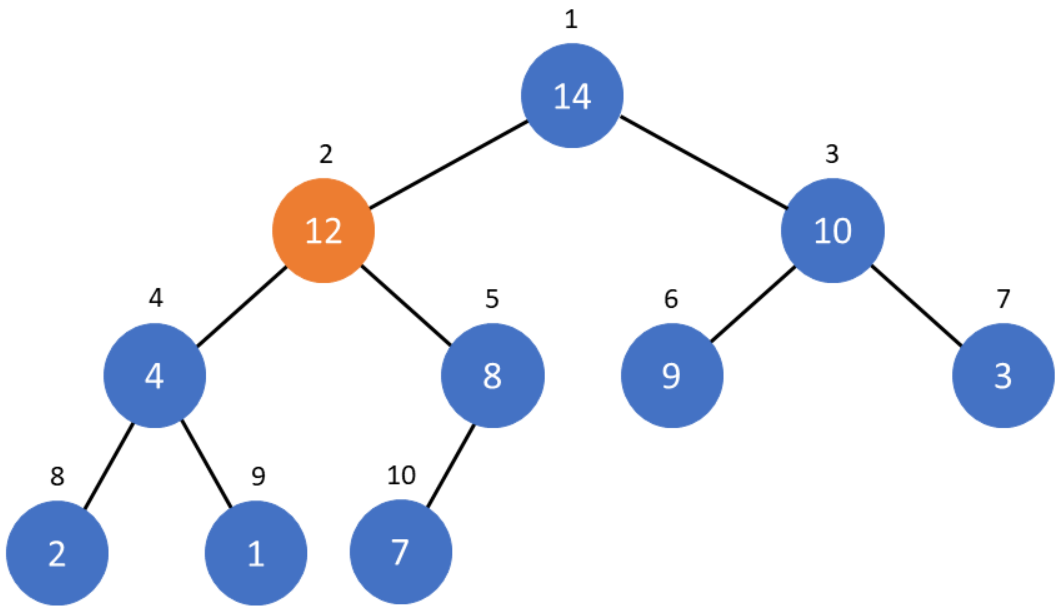
- Node indexed at 5 has a value higher than it's parent (node at index 2).
- Swap index 5 and index 2.



Example of Sift Up

Animation, please move to the next slide...

- Node indexed at 2 has a value higher than it's parent (node at index 1).
- No swap required.



Inserting an Item (`insert(x)`) - Sift Up

1. Insert item into next available node in the heap.
2. Sift the node up until in the correct position.

Please try inserting on Visualgo.net

This operation takes $O(log(n))$.

Why?

- In the worst-case we have to sift the new node all the way up to the root node.
- Height of complete binary tree with n nodes is $\lfloor log(n) \rfloor$.

Heapsort (not in-place)

- If you have a max-heap (min-heap) just keep extracting the max (min). This gives descending (ascending) order.

Time Complexity

- It takes $O(n)$ to create the heap.
- Extraction costs $O(log(n))$ and we do this n times.

$O(n) + O(n\log(n)) = O(n\log(n))$. Thus $O(\log(n))$. Note we can get a tighter bound.

Space complexity

We need to maintain a separate copy of the list. Thus $O(n)$.

Heapsort

This is the standard version.

- Sorts in-place
- Stable (preserves order of items that are equal)
- Worst-cast time complexity - $O(n \log(n))$
- Space complexity - $O(1)$

I have implemented this for you on the module. Doesn't require extra space.

Other Uses of Priority Queues

- Scheduling
- Event-driven simulation
- Bandwith management
- Efficient **Shortest Path** calculation

Summary

- Priority queue is a queue where items are ordered by priority
- We can use a binary heap to implement a priority queue

Operation	Worst-case Cost
`insert(x)`	$O(\log(n))$
`extract_max()`	$O(\log(n))$
`peek()`	$O(1)$

- A binary heap is a complete binary tree that satisfies the heap property
  - Either max-heap or min-heap
- Heapsort has wosrt-case time complexity $O(n\log(n))$ and space complexity $O(1)$

References

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. Introduction to algorithms. MIT press.